

A Two-Block Counterexample for the Benjamini–Hochberg Procedure under Correlated Two-Sided Gaussian Tests

Grounded DI LLC*

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Abstract

We give a two-block one-factor Gaussian construction for which the ordinary Benjamini–Hochberg procedure fails to control the false discovery rate at its nominal level. At $\alpha = 0.01$, an outward-rounded Arb certificate yields

$$\liminf_{N \rightarrow \infty} \text{FDR}_N \geq 0.01015553266461812931256738227976640877 \dots > 0.0101555 > 0.01.$$

A second certificate uses the same finite threshold brackets for every fixed $\alpha \in [0.00998, 0.01002]$ and proves

$$\liminf_{N \rightarrow \infty} \text{FDR}_{N,\alpha} \geq \alpha C, \quad C > 1.004440969427305667426958245856282226 \dots$$

The analytic reduction, the interval computations, the strict margins, and the small discrepancy between two grid starting indices were independently recomputed. This private technical note is authored by Grounded DI LLC and is not a peer-review or institutional endorsement statement.

1 Purpose and status

An accompanying three-block preprint establishes that ordinary Benjamini-Hochberg can exceed its nominal false discovery rate for correlated two-sided Gaussian tests. The present note records a distinct two-block construction and a separate reproducibility audit. It is not a replacement for that preprint. Grounded DI LLC is identified as the author of this two-block technical note under the attribution permission stated below.

The note has two mathematical purposes. First, it isolates the shortest factor model found in the audit: one null block and one nonnull block. Second, it makes the finite interval reduction explicit enough that every load-bearing numerical claim can be matched to an outward-rounded computation and a file hash.

2 The two-block model and main results

Let

$$Z, (\varepsilon_i)_{i \geq 1}, (\eta_j)_{j \geq 1}$$

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be mutually independent standard normal random variables. For each positive integer N , define $m_N = 100N$ coordinates by

$$X_i^{(0)} = \frac{1}{3}Z + \frac{2\sqrt{2}}{3}\varepsilon_i, \quad 1 \leq i \leq 99N, \quad (1)$$

$$X_j^{(1)} = \frac{15}{4} - \frac{1}{2}Z + \frac{\sqrt{3}}{2}\eta_j, \quad 1 \leq j \leq N. \quad (2)$$

The first block consists of true nulls and the second block consists of nonnulls. For every coordinate form the marginal two-sided Gaussian p -value

$$P = 2\bar{\Phi}(|X|),$$

where $\bar{\Phi}(x) = 1 - \Phi(x)$.

Proposition 1 (Validity of the Gaussian model). *The covariance matrix of (1)–(2) is positive definite and has unit diagonal. In particular, every true-null coordinate is marginally $N(0, 1)$ and its two-sided p -value is uniform on $[0, 1]$.*

Proof. Let a_N be the vector of factor loadings, equal to $1/3$ on the null block and $-1/2$ on the nonnull block. Let D_N be diagonal with entries $8/9$ on the null block and $3/4$ on the nonnull block. Then

$$\Sigma_N = a_N a_N^\top + D_N.$$

The diagonal matrix D_N is strictly positive, so Σ_N is positive definite. Moreover,

$$\left(\frac{1}{3}\right)^2 + \frac{8}{9} = 1, \quad \left(\frac{1}{2}\right)^2 + \frac{3}{4} = 1,$$

which gives unit marginal variances. The null means are zero. $\square$

The within-null correlation is $1/9$, the within-signal correlation is $1/4$, and the cross-block correlation is $-1/6$.

Theorem 1 (Point certificate at $\alpha = 0.01$). *Let FDR_N denote the false discovery rate of ordinary Benjamini–Hochberg at level $\alpha = 0.01$ in (1)–(2). Then*

$$\liminf_{N \rightarrow \infty} \text{FDR}_N \geq 0.01015553266461812931256738227976640877 \dots$$

Consequently, $\text{FDR}_N > 0.0101555 > 0.01$ for all sufficiently large N .

Theorem 2 (A nontrivial interval of nominal levels). *For every fixed*

$$\alpha \in [0.00998, 0.01002],$$

the same model satisfies

$$\liminf_{N \rightarrow \infty} \text{FDR}_{N,\alpha} \geq \alpha C,$$

where

$$C > 1.00444096942730566742695824585628222635 \dots > 1.$$

Thus, for each fixed α in the displayed interval, there exists an $N_0(\alpha)$ such that $\text{FDR}_{N,\alpha} > \alpha$ for all $N \geq N_0(\alpha)$.

Remark 1. The final quantifier is pointwise in α . The certificate does not claim one common finite N_0 that works simultaneously for every $\alpha \in [0.00998, 0.01002]$.

3 Conditional limiting distributions

For $c \geq 0$, define

$$u(c) = 2\bar{\Phi}(c).$$

Then $P \leq u(c)$ exactly when $|X| \geq c$. For $a \geq 0$ and $s > 0$, define

$$Q(c; a, s) = \bar{\Phi}\left(\frac{c-a}{s}\right) + \bar{\Phi}\left(\frac{c+a}{s}\right).$$

If $Y \sim N(m, s^2)$, then

$$\Pr(|Y| \geq c) = Q(c; |m|, s).$$

Conditional on $Z = z$, the two block laws are

$$X^{(0)} \mid Z = z \sim N\left(\frac{z}{3}, \frac{8}{9}\right), \quad X^{(1)} \mid Z = z \sim N\left(\frac{15}{4} - \frac{z}{2}, \frac{3}{4}\right).$$

Write

$$s_0 = \frac{2\sqrt{2}}{3}, \quad s_1 = \frac{\sqrt{3}}{2}.$$

The conditional limiting CDF of all p -values therefore satisfies

$$G_z(u(c)) = 0.99 Q\left(c; \frac{|z|}{3}, s_0\right) + 0.01 Q\left(c; \left|\frac{15}{4} - \frac{z}{2}\right|, s_1\right). \quad (3)$$

The conditional true-null CDF is

$$F_{0,z}(u(c)) = Q\left(c; \frac{|z|}{3}, s_0\right). \quad (4)$$

Two monotonicity facts drive the finite reduction:

$$\begin{aligned} \frac{\partial}{\partial c} Q(c; a, s) &= -\frac{1}{s} \left[\phi\left(\frac{c-a}{s}\right) + \phi\left(\frac{c+a}{s}\right) \right] < 0, \\ \frac{\partial}{\partial a} Q(c; a, s) &= \frac{1}{s} \left[\phi\left(\frac{c-a}{s}\right) - \phi\left(\frac{c+a}{s}\right) \right] \geq 0. \end{aligned}$$

Thus Q decreases in c and increases in the absolute conditional mean.

Conditional on $Z = z$, observations are independent and identically distributed within each block. The Glivenko–Cantelli theorem applied blockwise gives, almost surely,

$$\|\hat{G}_N - G_z\|_\infty \rightarrow 0, \quad \|\hat{F}_{0,N} - F_{0,z}\|_\infty \rightarrow 0.$$

The two conditional limits are continuous because both residual variances are strictly positive.

4 BH threshold bracketing

Let

$$\mathcal{T}_N = \left\{ \frac{\alpha r}{m_N} : 1 \leq r \leq m_N \right\}$$

and write the BH threshold as

$$\tau_N = \max \left(\left\{ t \in \mathcal{T}_N : \hat{G}_N(t) \geq \frac{t}{\alpha} \right\} \cup \{0\} \right).$$

Lemma 1 (Two strict signs bracket the threshold). *Suppose $\hat{G}_N \rightarrow G$ uniformly, where G is continuous. Let $0 < v \leq w \leq \alpha$. If*

$$G(t) < \frac{t}{\alpha} \quad \text{for all } t \in [w, \alpha], \quad G(v) > \frac{v}{\alpha},$$

then

$$\limsup_{N \rightarrow \infty} \tau_N \leq w, \quad \liminf_{N \rightarrow \infty} \tau_N \geq v.$$

Proof. The first strict inequality has a uniform negative margin on the compact interval $[w, \alpha]$. Uniform convergence excludes all sufficiently large BH grid points in that interval. The second strict inequality persists on a neighborhood of v . Since the BH grid mesh tends to zero, a feasible grid point approaches v from that neighborhood. $\square$

Lemma 2 (Threshold bracket gives an FDP lower bound). *Assume Lemma 1, let the null proportion be π_0 , and suppose $\hat{F}_{0,N} \rightarrow F_0$ uniformly with F_0 continuous. Then*

$$\liminf_{N \rightarrow \infty} \text{FDP}_N \geq \frac{\alpha \pi_0 F_0(v)}{w}.$$

Proof. Eventually $\tau_N > 0$ and the BH identity gives

$$R_N = \frac{m_N \tau_N}{\alpha}, \quad V_N = \pi_0 m_N \hat{F}_{0,N}(\tau_N).$$

Therefore

$$\text{FDP}_N = \frac{\alpha \pi_0 \hat{F}_{0,N}(\tau_N)}{\tau_N}.$$

Use $\liminf \tau_N \geq v$, $\limsup \tau_N \leq w$, monotonicity of empirical CDFs, uniform convergence, and continuity of F_0 . $\square$

5 Finite rectangle certificate

Partition the latent interval $[-5, 5]$ into one thousand bins

$$B_k = \left[\frac{k}{100}, \frac{k+1}{100} \right], \quad -500 \leq k \leq 499,$$

and use the rational Gaussian-threshold grid

$$c_j = \frac{j}{1000}.$$

For $g \in \{0, 1\}$, let $m_{g,k}^-$ and $m_{g,k}^+$ be the exact minimum and maximum of the absolute conditional mean in block g over B_k . Since the means are affine in z , these extrema occur at endpoints or at a zero of the affine function.

For a fixed α , define the uniform upper bound

$$U_{j,k}(\alpha) = 0.99 Q(c_j; m_{0,k}^+, s_0) + 0.01 Q(c_j; m_{1,k}^+, s_1) - \frac{u(c_{j+1})}{\alpha}, \quad (5)$$

and the pointwise lower bound

$$L_{j,k}(\alpha) = 0.99 Q(c_j; m_{0,k}^-, s_0) + 0.01 Q(c_j; m_{1,k}^-, s_1) - \frac{u(c_j)}{\alpha}. \quad (6)$$

If $U_{j,k}(\alpha) < 0$, monotonicity implies

$$G_z(u(c)) - \frac{u(c)}{\alpha} < 0 \quad \text{for all } (z, c) \in B_k \times [c_j, c_{j+1}].$$

Let j_k be the first index for which strict negativity is not certified and put $a_k = c_{j_k}$. Let $\ell_k \geq j_k$ be the first index for which $L_{\ell_k,k}(\alpha) > 0$ and put $b_k = c_{\ell_k}$. Then, uniformly for $z \in B_k$,

$$G_z(t) < \frac{t}{\alpha} \quad \text{for } t \in [u(a_k), \alpha], \quad G_z(u(b_k)) > \frac{u(b_k)}{\alpha}.$$

Lemmas 1–2, with $\pi_0 = 0.99$, give

$$\liminf_{N \rightarrow \infty} \text{FDP}_N \geq d_k(\alpha) := \frac{\alpha(0.99) Q(b_k; m_{0,k}^-, s_0)}{u(a_k)} \quad \text{on } \{Z \in B_k\}. \quad (7)$$

The point certificate evaluates (5)–(7) at $\alpha = 0.01$. The interval certificate checks upper infeasibility at $\alpha_{\text{high}} = 0.01002$, where it is hardest, and lower feasibility at $\alpha_{\text{low}} = 0.00998$, where it is hardest. Hence the same (a_k, b_k) brackets work for every α in the interval. In that case,

$$d_k(\alpha) = \alpha C_k, \quad C_k = \frac{0.99 Q(b_k; m_{0,k}^-, s_0)}{u(a_k)},$$

and $\sum_k C_k \Pr(Z \in B_k)$ is independent of α .

6 Outward-rounded numerical results

All subdivision endpoints, model parameters, weights, and grid points are entered as exact rationals. Square roots and Gaussian tails are evaluated as Arb balls. A strict sign is accepted only when the entire computed ball lies strictly on the required side of zero.

Quantity	Point certificate	Alpha-interval certificate
Nominal level(s)	0.01	[0.00998, 0.01002]
Certified aggregate	0.0101555326646181293...	1.0044409694273056674...
Arb radius	5.11×10^{-48}	5.46×10^{-46}
Smallest strict margin	$7.1210085605674389704 \times 10^{-16}$	$1.7018061089938611757 \times 10^{-15}$
Radius at smallest margin	3.89×10^{-58}	3.98×10^{-58}
Margins below 10^{-12}	159	158
Margins below 10^{-14}	5	6
Margins below 10^{-15}	1	0

Table 1: Independent reproduction summary. The first aggregate is an FDR lower bound; the second is the coefficient C in Theorem 2.

For the point certificate, the ten unit-bin contributions over $[-5, 5]$ are:

Range of Z	Certified contribution
$[-5, -4]$	0.0000115641299609205509669
$[-4, -3]$	0.0001910406736692213611250
$[-3, -2]$	0.0010372328000280610818103
$[-2, -1]$	0.0019770493938734550768221
$[-1, 0]$	0.0015473902811503407120878
$[0, 1]$	0.0014207003268786251786284
$[1, 2]$	0.0019179247950068331737665
$[2, 3]$	0.0015063175588749091913413
$[3, 4]$	0.0005183429146171462370210
$[4, 5]$	0.0000279697905586167489979

The point program begins its c -grid at $j = 2575$, while the interval program begins at $j = 2573$. An independent verifier checked both starts directly. The two additional rectangles at $j = 2573$ and $j = 2574$ are strictly negative in every latent bin; all first threshold brackets occur above 2575; and no bracket, contribution, or final aggregate changes. The discrepancy is therefore cosmetic.

7 Completion of the proofs

On the probability-one event where the conditional empirical CDFs converge, equation (7) holds on every latent bin. Since $0 \leq \text{FDP}_N \leq 1$, Fatou’s lemma gives

$$\begin{aligned}
\liminf_{N \rightarrow \infty} \text{FDR}_N &= \liminf_{N \rightarrow \infty} \mathbb{E}[\text{FDP}_N] \\
&\geq \mathbb{E} \left[\liminf_{N \rightarrow \infty} \text{FDP}_N \right] \\
&\geq \sum_{k=-500}^{499} d_k(\alpha) \Pr(Z \in B_k).
\end{aligned}$$

The omitted region $|Z| > 5$ contributes a nonnegative amount. The point aggregate in Table 1 proves Theorem 1. For the alpha interval, $d_k(\alpha) = \alpha C_k$ and the second aggregate in Table 1 proves Theorem 2.

8 Reproducibility manifest

The independently recomputed artifacts have the following SHA-256 hashes:

Artifact	SHA-256
Point certificate program	d8e5b36c17ce5ca7a6c5cfb2a2c6ff5a12c93d339509b0e73cbef1f1bb8d5d67
Alpha-interval certificate program	3f1a5570198544a823472b8c8e9b50666cb8736182963d013bae17f14cebfafd
Independent verifier	f7af40597d738b8bc167e5349c0b883751eeaf09a1be74c313f7da9a5af11d70
Machine-readable verifier report	0767f33921feaa5a6689256e156f3ad70d11e5539ef5a611ae216bc910e6c289
Verifier summary	9b2120abaf56626700f5dd1680b464f5b42127e0f463c6fc7a19fa1206fc8101

The reproduction used Python 3.13.5, `python-flint` 0.8.0, 50 decimal digits in the independent verifier, and 16 workers. The two original certificate programs used 40 decimal digits.

9 Scope, limitations, and attribution

This note establishes an asymptotic counterexample and a fixed-alpha interval of asymptotic counterexamples. It does not provide an explicit finite value of N_0 , a common N_0 uniform over the alpha interval, a journal referee decision, or a claim of priority over the accompanying preprint.

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